Name of the Experiment: STUDY ON CORRELATION & REGRESSION TEST

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Problem-1: The following table gives the ages and blood pressure of 12 women.

Ages in years (x)	56	42	36	47	49	42	72	63	55	60	62	65
Blood pressure (y)	147	118	118	128	145	140	155	160	149	150	165	170

(i) Draw a scatter diagram

(ii) Find correlation co-efficient between x and y and comment

(iii) also test the correlation co-efficient

Solution:

Here,

Independent value of (x) = ages in years

Dependent value (y) = blood pressure

(i) Scatter diagram in drawn

(ii) $r = \frac{Cov(x,y)}{\sqrt{var(x).var(y)}}$

$= \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sqrt{\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)\left(\sum_{i=1}^{n} Y_i^2 - n\bar{y}\right)^2}}$

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$$= \frac{96446 - (12 \times 54.08 \times 146)}{\sqrt{\{36917 - (12 \times 54.08^2)\}\{258618 - (12 \times 146^2)\}}}$$

$$= \frac{94748.16}{\sqrt{1321.24 \times 2826}} = 0.88$$

(ii) Here, null hypothesis, $H_0 : \rho = 0$

Alternate hypothesis, $H_1 : \rho \neq 0$

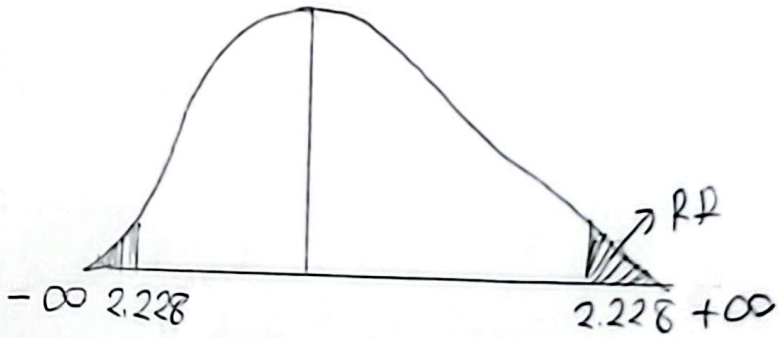
And, $r = 0.88$

$n = 12$

Test statistics $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t$ dist^n with $(n-1)$ d.f

$$= \frac{0.88\sqrt{12-1}}{\sqrt{1-(0.88)^2}} = 5.86$$

The tabulated value for two tailed t dist^n at 5% level of significance with 10 dif, $t_{0.05,10} = 2.228$



Comment: As the calculated value on the rejected region at 5% level of significance with 10 dif. So the null hypothesis may be accepted.

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Problem-02: A study was made on the amount of converted sugar in a certain process at various temperatures. The data were coded and recorded below-

Temperature (x)	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
Converted sugar (y)	8.2	7.8	8.5	9.8	9.5	8.9	6.8	10.2	9.3	9.2	10.5

(i) Estimate the linear regression line of y on x

(ii) Estimate the amount of converted sugar produced when the coded temperature is 1.75

(iii) Test the significance of the regn co-efficient.

Solution: (i) Simple linear regression model is

$$Y_i = \alpha + \beta x_i + \epsilon_i$$

Where, Y_i = Dependent variable, x_i = Independent variable

α = Intercept term, β = Regression co-efficient

ϵ_i = Random error

Assumption:

(a) The x's are fixed

(b) The relation of y on x is linear

(c) ϵ_i's $\sim$ NID $(0, \sigma^2)$

(d) ϵ_i's are uncorrected

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(e) The conditional distribution of y for each value of x are normal with equal variances

By using OLS, we get, $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}_i$

We know,

$$\hat{\beta} = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}$$

$$= \frac{149.71 - 1.5 \times 8.96 \times 11}{25.85 - 11 \times 1.5^2} = 1.7$$

By using calculator,

$\bar{x} = 1.5$

$\bar{y} = 8.96$

$n = 11$

$\sum x_i^2 = 25.85$

$\sum x_i y_i = 149.71$

$$\hat{\alpha} = \bar{y}_i - \hat{\beta}\bar{x}_i$$

$$= 8.96 - 1.7 \times 1.5$$

$$= 6.41$$

Now, $\bar{y}_i = \hat{\alpha} + \hat{\beta}\bar{x}_i$

$= 6.41 + 1.7x_i$

ii) Here, Coded temperature, $\bar{x}_i = 1.75$

From (i) we get, the linear regression line is,

$$\hat{y}_i = 6.41 + 1.7x_i$$

$$= 6.41 + 1.7 \times 1.75$$

$$= 9.385$$

So, the amount of converted sugar produced in 8.28 where the coded temperature is 1.75

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(iii) For β, Null hypothesis, $H_0 : \beta = \beta_0$

Alternate hypothesis, $H_0 : \beta \neq \beta_0$

Here, $\beta_0 = 0$

Test hypothesis, $t = \dfrac{\hat{\beta} - \beta_0}{\hat{\alpha}/\sqrt{\Sigma(x_i - \bar{x})^2}} \sim t$ with $(n-2)$ d.f

$$\hat{\alpha} = \frac{1}{n-2}\left[SS(y) - \hat{\beta}\, Sp(x,y)\right]$$

$$= \frac{1}{11-2}\left[12.76 - 1.7 \times 1.87\right]$$

$$\hat{\alpha} = 1.03$$

$SS(y) = \Sigma y_i^2 - n\bar{y}^2$

$= 895.86 - 11 \times (8.96)^2$

$= 12.76$

$Sp(x,y) = \Sigma x_i y_i - n\bar{x}\bar{y}$

$= 149.71 - 11 \times 1.5 \times 8.96$

$= 1.87$

$\Sigma(x_i - \bar{x})^2 = \Sigma x_i^2 - n\bar{x}^2$

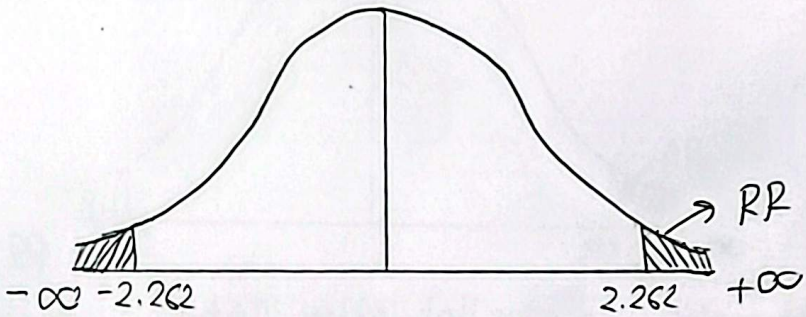
$= 25.85 - 11 \times 1.5^2 = 1.1$

$$t = \frac{1.7 - 0}{\frac{1.03}{\sqrt{1.1}}} \sim t \text{ with } (11-2) \text{ d.f}$$

$= 1.73 \sim t$ with 9 dif.

The tabulated value of t at 2.5% level of significance with 9 d.f.

$-t_{-0.025,9} = -2.262$ & $t_{0.025,9} = 2.262$



Comment: As the calculated value falls on the accepted region at 2.5% level of signifance with 9 d.f. So, the null hypothesis may be accepted.

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For $\hat{\alpha}$, Null hypothesis, $H_0 : \alpha = \alpha_0$

Alternate hypothesis, $H_1 : \alpha = \alpha_0$

Here, $\alpha_0 = 0$, level of significance, $\alpha = 2.5\% = 0.025$

$\hat{\alpha} = 1.065$

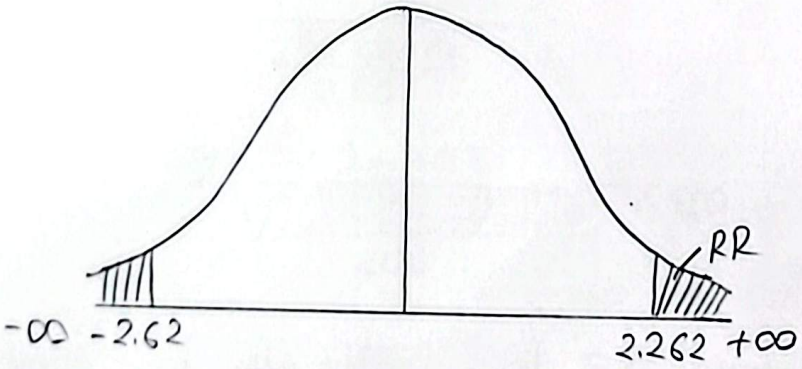
Test hypothesis, $t = \dfrac{\hat{\alpha} - \alpha_0}{\sqrt{\dfrac{\hat{\alpha}^2 - \sum x_i^2}{n \sum (x_i - \bar{x})^2}}} \sim t$ with 9 d.f

$= \dfrac{6.41 - 0}{\sqrt{\dfrac{1.065 \times 25.85}{11 \times 1.1}}}$

$= \dfrac{6.41}{1.5} = 4.25 \sim t$ with 9 d.f

Here,
$\hat{\alpha} = 6.41$
$\sum x_i^2 = 25.85$
$n = 11$
$\sum (x_i - \bar{x})^2 = 1.1$

The tabulated value of t distribution at 2.5% level of significance with 9 d.f is $-t_{0.025,9} = -2.622$; $t_{0.025,9} = 2.262$



Comment: As the calculated value falls on the rejected region at 2.5% level of significance with 9 d.f. We may reject the null hypothesis.

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